

Measuring Judicial Productivity in Spanish Instruction Courts: A Malmquist Productivity Index Approach

By Giuseppe Tomaselli

Universitat Autònoma de Barcelona

Universidad Pública de Navarra

Universitat de les Illes Balears

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Supervisor: **Pablo Arocena**

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1. Introduction

The efficiency of the judicial system represents one of the central topics in the economic and policy debate of recent decades, as a demonstration of how capable a State is of guaranteeing justice and consequently equity towards its citizens. It also represents a determinant of the country's economic competitiveness, given that slow justice increases transaction costs, discourages foreign investment and reduces trust in institutions. In this sense, courts become a fundamental component of a country's institutional infrastructure, whose functioning conditions its economic activity.

In this context, the case of Spain proves to be of particular interest. On the one hand, the country is positioned around the OECD average for the duration of civil proceedings, with 778 total days against an average of 788, as reported by the study of Palumbo et al. (2013). On the other hand, despite a duration of proceedings perfectly in line with the average, the Spanish judicial system is characterised by high internal heterogeneity, both in terms of workload and in terms of available resources, with significant differences across the 17 autonomous communities. This heterogeneity makes Spain a particularly suitable context for comparative analysis of court efficiency, as it allows productivity variations to be observed across a large and diversified sample, within a broader homogeneous institutional framework.

The *Juzgados de Instrucción* are the criminal courts of first instruction, responsible for the investigative phase of ordinary criminal proceedings. They are highly relevant within the Spanish judicial system, as one of its pillars in terms of volume of cases handled, and therefore their efficiency has direct implications for the overall duration of criminal proceedings, whose starting point is precisely the instruction phase. However, despite their relevance, this type of court has been little studied from a quantitative perspective, probably due to the scarcity of homogeneous and comparable data at the district level over an extended period of time.

In this thesis, the period 2016-2023 will be analysed, as it represents a phase whose stability is disrupted by the COVID-19 pandemic, an unprecedented shock for the judicial system. The pandemic forced courts to suspend or drastically reduce their activity during 2020, with effects that extended into subsequent years through the accumulation of backlogs and the slow recovery of normal working rhythms. Moreover, in the periods that followed, courts had to deal not only with the proceedings left unmanaged during the lockdown, but also with the adoption

of new ways of holding hearings. Consequently, the health crisis represented not only an immediate negative shock, but potentially also a push towards longer-term organisational changes, whose impact remains difficult to fully assess within the time horizon considered here.

Measuring judicial efficiency requires tools suited to the nature of the public sector. Unlike private firms, courts do not operate in a competitive market and have no direct incentives to maximise output, which makes traditional performance indicators, such as average case resolution times, useful but insufficient to capture relative efficiency differences between units. The literature has responded to this need by developing approaches based on the production frontier, including DEA and SFA, which allow the relative efficiency of decision-making units to be compared without imposing strong assumptions on the technological structure. A common limitation of many of these studies, however, is the static nature of the analysis: most applications estimate efficiency in a single year, or compare two isolated periods, without capturing the dynamic evolution of productivity over time. The MPI-based approach overcomes this limitation, allowing productivity changes to be analysed across a sequence of consecutive periods and distinguishing between individual efficiency recovery and advancement of the production frontier, two phenomena with very different policy implications.

Despite growing attention to the topic of judicial efficiency, the literature specifically on Spanish courts remains relatively limited. Existing studies focus on narrow geographical or jurisdictional contexts: Pedraja-Chaparro and Salinas-Jiménez (1996) analyse the High Courts of Justice using data from 1991; Espasa and Esteller-Moré (2015) focus on civil first-instance courts in Catalonia; Martín-Román, Moral and Rosales (2025) study labour courts at the national level. None of these studies offers a dynamic productivity analysis at the national level for the *Juzgados de Instrucción*, across a large sample and a time horizon that includes the effects of the pandemic.

This work aims to fill, at least in part, this gap. The objective is to analyse the productivity change of the 80 *Juzgados de Instrucción* in the Spanish sample over the period 2016-2023, with a twofold focus. First, the temporal evolution of productivity is analysed, including the impact of the pandemic, and subsequently the geographical differences between autonomous communities and between individual courts. The tool adopted is the Malmquist Productivity Index (MPI), which allows productivity changes between two consecutive periods to be measured and decomposed into two components: efficiency change (catch-up) (EC), which

measures whether a court has moved closer to or further from the production frontier, and technical change (frontier shift) (TC), which measures whether the frontier itself has shifted outward or inward.

The MPI is calculated through Data Envelopment Analysis (DEA), a non-parametric method that constructs the production frontier from the input-output combinations observed in the sample, without requiring assumptions about the functional form of the technology. In contexts such as the judicial one, where the decision-making units, in this case the courts, are heterogeneous in size and composition of the workload, DEA proves more suitable than parametric approaches such as Stochastic Frontier Analysis (SFA), also because the latter imposes a predefined functional form on the relationship between inputs and outputs, which in this case would be difficult to justify. Furthermore, while most judicial efficiency studies rely on static efficiency measures in a single year, the MPI-based approach allows the dynamic dimension of productivity to be captured, distinguishing between individual efficiency recovery and advancement of the production frontier.

The sample is therefore composed of 80 *Juzgados de Instrucción*, distributed across 17 autonomous communities and observed annually from 2016 to 2023, for a total of 640 observations and seven consecutive transitions. The input variables used are judges, administrative staff, managers and other-category staff, all expressed as full-time equivalents (FTE). The output variable is the number of proceedings resolved in the year. The data come from official administrative sources of the Spanish judicial system.

The rest of the work is structured as follows. Chapter 2 presents a review of the literature on methods for measuring court efficiency, with particular attention to applications in the European and Spanish context. Chapter 3 describes the data used and the methodology adopted, illustrating the theoretical framework of the MPI and the calculation procedure through DEA. Chapter 4 presents the results, organised into five sections: analysis by period, by COVID phases, by region, by district and by court size. Chapter 5 concludes with a summary of the main results, a discussion of policy implications, the limitations of the study and possible directions for future research.

2. Literature Review

2.1 Why measure the efficiency of courts

The efficiency of courts, represented by the quality and speed of justice, is increasingly recognized as a determinant of a country's socio-economic and institutional context, since slow and inefficient justice entails higher transaction costs, deters investment, and reduces citizens' trust in the state.

Past literature offers several insights for identifying the determinants of judicial efficiency mentioned above. Possible determinants that have been highlighted include, for example, the number of judges per inhabitant, career incentives, staffing and technology endowment, internal organization, budget and the level of vacant positions, and case complexity, but it remains important to stress that the empirical literature on these factors is still fragmented and unable to reach unambiguous conclusions (Voigt, 2016).

Another approach is the one adopted by Palumbo, Giupponi, Nunziata and Mora-Sanguinetti (2013) for the OECD, who organize the concept of judicial efficiency along three dimensions: length of proceedings, accessibility, and predictability of decisions; and around a demand-supply approach, according to which case backlogs and the time taken by justice depend both on supply factors, namely court resources and organization, and on demand factors, such as litigation propensity and the quality of regulation. The study is based on a sample of 35 OECD countries and shows that results, in terms of resolution times for civil disputes, are highly heterogeneous. Among the results across countries, we find an average of 788 days, calculated from the first instance to the eventual appeal to the last instance. Specifically, some countries are positioned practically at the average, such as Spain (778 days) or common law countries (777 days), while others fall well below the average, such as Japan or Switzerland, both at 368 days, or Germany (394 days); but there are also countries that lie far above the average, such as Italy, which shows the highest result in the sample, with 2,866 total days (almost eight years), followed by Slovenia, with 1,354 days, and by countries of French legal tradition (1,307 days, which includes France with its 950 days).

2.2 Methodological approaches

The methods used in the literature to study the measurement of judicial performance are organized around frontier approaches (DEA and SFA) and non-frontier econometric approaches.

Data Envelopment Analysis (DEA) is a non-parametric method that constructs an efficient frontier from the input-output combinations observed in the sample, without requiring assumptions on the functional form of the technology or data on factor prices. It is the most widely used method in the literature, as reported by the systematic review of Pereira, Bădin, Kerstens and Silva (2025), which finds that, among 42 European studies out of 50 total published between 1992 and 2023 (selected from over 6,500 papers reviewed) that employ a quantitative method, 84% use non-parametric frontier approaches. In many cases, DEA is used in a two-stage approach: in the first stage, the efficiency scores of each unit are estimated, and in the second stage these scores are regressed on contextual variables to identify their determinants.

Other examples of studies applying DEA can be found in the papers by Pedraja-Chaparro and Salinas-Jiménez (1996), who provide one of the first applications of the technique to European courts by measuring the technical efficiency of the Administrative Litigation Division of the 21 Spanish High Courts; Ippoliti and Tria (2020), who analyse 164 Italian first-instance courts; and Schneider (2005), who applies it to nine German Labor Appeal Courts (Landesarbeitsgerichte) to construct a productivity measure alongside a second performance indicator, the appeal confirmation rate of rulings.

Another parametric method used is Stochastic Frontier Analysis (SFA), a parametric method that, unlike DEA, requires specifying a functional form for the frontier, but which at the same time allows the distance from the frontier to be decomposed into an inefficiency component and a pure random error component, thus making it possible to distinguish deviations due to unobservable factors, such as workload congestion, from deviations due to inefficient use of resources. A study applying SFA is that of Espasa and Esteller-Moré (2015), who apply a fixed-effects panel approach to first-instance civil courts in Catalonia over the period 2005-2013, finding an average efficiency of 97.82%, increasing over time and with decreasing dispersion (Espasa and Esteller-Moré, 2015).

A further approach, distinct from frontier methods, uses direct econometric techniques, such as OLS regression models, panel models, and dose-response models, to estimate the effect of specific organizational variables on court productivity, without estimating an efficient frontier. A study using this type of model is that of Martín-Román, Moral and Rosales (2025), who employ least squares, panel estimates, and dose-response models to analyse the effect of substitute judges on the productivity of Spanish labor courts (Juzgados de lo Social).

Finally, there is the method used by the OECD study of Palumbo et al. (2013), which adopts a cross-country approach, based on descriptive statistics and regression models on data aggregated at the country level. This model differs from those described so far, as it is useful for international benchmarking, but not for detecting efficiency scores for individual courts.

2.3 Inputs, outputs, and specification issues

One of the most delicate methodological problems in measuring court efficiency is the choice of input and output variables. As for inputs, most studies identify them as the number of judges, administrative staff, and, at times, the available budget or technological resources. On the output side, by contrast, the most common measure is the number of cases resolved, often represented by a Clearance Rate, calculated as the ratio between cases resolved and total caseload (including the backlog).

It is important to mention the paper by Ippoliti and Tria (2020), as it brings out a problem that proves to be central: the degree of output aggregation by subject matter. The two authors compare four different specifications of the DEA model applied to Italian first-instance courts, showing that disaggregating output by type of proceeding (civil, criminal, etc.) is statistically superior to an aggregate measure. In fact, using an aggregate output, 38.30% of courts are misclassified relative to the disaggregated model, and the efficiency gap between Northern and Southern Italy is significantly reduced when the output is disaggregated (Ippoliti and Tria, 2020).

The result just described carries a certain relevance for policy purposes, since efficiency-improvement policies based on an aggregate productivity measure might fail to capture procedural differences between the various types of cases, such as, for example, the longer

average duration of civil proceedings compared to criminal ones, which instead underlie much of the territorial gap observed between Northern and Southern Italy.

Another aspect to consider is the performance measure, which can vary considerably. In most DEA/SFA studies, a single measure of technical efficiency is used, whereas Schneider (2005) proposes two, distinct and potentially conflicting, measures: a productivity score (DEA) and an appeal confirmation rate of rulings. For example, courts with a higher share of judges holding a doctorate are more productive in terms of DEA score, but their rulings are confirmed less often on appeal by the Federal Labor Court (Schneider, 2005). The author rules out that this reflects more frequent errors, suggesting instead that these judges write more innovative rulings, which depart more often from precedent or from the positions of the higher court (Schneider, 2005).

A further conceptual complication is raised by the contribution of Espasa and Esteller-Moré (2015), who distinguish between inefficiency in the strict sense and congestion. The reason for this distinction lies in the fact that low observed productivity can stem either from an inefficient use of available resources or simply from an excessive workload relative to the court's capacity (Espasa, Esteller-Moré, 2015), a phenomenon that a well-specified frontier model should keep distinct.

Regarding the OECD study by Palumbo et al. (2013), on the other hand, it can be said that the measurement logic adopted is radically different, as it does not require the specification of inputs and outputs at the level of the individual court, since performance is indicated through indicators of proceeding length, accessibility, and predictability of the national judicial system. This approach avoids the specification problems typical of DEA, but sacrifices the possibility of identifying determinants at the level of the individual court.

2.4 Main empirical findings

The main empirical findings emerging from the literature described so far, summarized below in Table 2.1, converge in attributing considerable weight to the characteristics of judicial staff. The stability of judges over time is associated with greater efficiency, while high turnover tends to depress productivity (Pereira et al., 2025). This result is also found at a more specific, micro level. The presence of temporary, non-career judges is likewise associated with lower efficiency

in Catalonia (Espasa & Esteller-Moré, 2015), just as court productivity decreases when the titular judge is supported by non-professional substitutes, while no effect emerges when the substitutes are career judges, in Spanish labor courts (Martín-Román, Moral & Rosales, 2025). Furthermore, it emerges that judges' individual characteristics affect the relationship between productivity and the quality of decisions in subtler ways, as shown by Schneider's (2005) study on German Labor Appeal Courts. Judges with a doctorate are in fact more productive, but their rulings are confirmed less often on appeal. A different but equally relevant effect occurs with judges who have a higher *ex ante* probability of promotion, who are at the same time less productive and less often confirmed (Schneider, 2005).

A second group of factors concerns the organization and technological endowment of courts. The adoption of computerized case management systems and mergers between small courts, of which the example of the Swedish reforms remains the most cited, are among the factors most consistently associated with greater efficiency, while the debate on economies of scale remains open, with some authors supporting their existence in larger courts and others finding no robust evidence in that regard (Pereira et al., 2025). In this respect, the study by Ippoliti and Tria (2020) offers a useful indication of the limits of aggregate measures, showing how the efficiency difference between Northern and Southern courts changes drastically, decreasing significantly when the output is disaggregated by subject matter. This suggests that part of the gap observed at the aggregate level depends on the different composition of the workload, rather than on an actual difference in efficiency between courts.

Finally, as regards efficiency levels, the available estimates show fairly wide variability across countries. Voigt's (2016) review reports average scores ranging from 59% in Brazil up to 88.84% in Germany (Schneider, 2005). For Spain specifically, Pedraja-Chaparro and Salinas-Jiménez (1996) provide an early benchmark, finding a mean technical efficiency of 77.38% for the Administrative Litigation Division of Spanish High Courts in 1991. More recent estimates, however, paint a different picture. Espasa and Esteller-Moré (2015) calculate an average efficiency of 97.82% for Catalan courts between 2005 and 2013, increasing over time, suggesting a substantial improvement over the intervening decades. Moreover, the OECD study by Palumbo et al. (2013) places Spain almost exactly at the average of the 35 countries considered in terms of overall length of civil proceedings, at 778 days against an OECD average of 788, with times of 272 days at first instance, 189 on appeal, and 316 at the last instance.

Table 2.1 - Summary of the main empirical studies

Author(s)	Year	Country / scope	Method	Main result
Pedraja-Chaparro & Salinas-Jiménez	1996	Spain (21 High Courts - Administrative Litigation Division)	DEA (CCR model, weight restrictions)	Mean efficiency 77.38%; 5 of 21 courts efficient; avoidable delays calculated
Ippoliti & Tria	2020	Italy (164 first-instance courts)	DEA (4 model specifications)	Disaggregating output by subject matter is statistically superior and reduces the North-South gap
Espasa & Esteller-Moré	2015	Spain - Catalonia (first-instance civil courts)	Fixed-effects panel SFA (Wang & Ho, 2010)	Average efficiency 97.82%, increasing over time; temporary judges reduce efficiency
Martín-Román, Moral & Rosales	2025	Spain (labor courts)	OLS, panel, dose-response models	Productivity decreases with non-professional substitutes; no effect with career substitutes
Schneider	2005	Germany (9 Labor Appeal Courts)	DEA + appeal confirmation rate	Judges with a doctorate more productive but confirmed less often on appeal
Palumbo, Giupponi, Nunziata & Mora-Sanguinetti (OECD)	2013	35 OECD countries	Cross-country indicators (length, accessibility, predictability)	Spain close to the OECD average (778 vs 788 days); Italy an extreme case (2,866 days)

2.5 Empirical evidence on Spain

The earliest study applying DEA to Spanish courts is that of Pedraja-Chaparro and Salinas-Jiménez (1996), who measure the technical efficiency of the Administrative Litigation Division of the 21 Spanish High Courts (Tribunales Superiores de Justicia) using data from 1991. The paper employs the CCR model with homogeneity tests and restrictions on input and output weights to ensure comparability across courts of different sizes. The results indicate that only 5 of the 21 courts are technically efficient, with a mean efficiency score of 77.38%, pointing to considerable scope for improvement. The authors extend the analysis by computing avoidable delays derived from DEA targets and by approximating allocative and global efficiency, finding that technical and allocative efficiency do not necessarily coincide in a public-sector context where cost minimization may conflict with broader service objectives. It is important to note that this study differs from the present thesis in two respects: it focuses on administrative litigation rather than criminal instruction, and it relies on a single year of cross-sectional data rather than a panel. It, however, represents a key methodological reference point for the Spanish judicial system.

The literature specifically analysing the efficiency of Spanish courts is still relatively limited and concentrated on narrow territorial or jurisdictional contexts. Among the literature reviewed, there is the study by Espasa and Esteller-Moré (2015), which focuses on first-instance civil courts in Catalonia, applying a panel SFA model that allows inefficiency due to workload congestion to be distinguished. The study's results show an increase in average efficiency over the period 2005-2013, accompanied by a reduction in dispersion across courts, and a negative effect from the use of temporary judges (Espasa and Esteller-Moré, 2015). A second study, conducted by Martín-Román, Moral and Rosales (2025), instead analyses labor courts (Juzgados de lo Social) at the national level between 2005 and 2019, focusing on the effect of substitute judges on productivity, which they measure through a clearance rate. This latter study uses an econometric approach, not based on DEA or SFA, which allows the causal effect of judicial substitution to be isolated. However, it is not able to provide a measure of technical efficiency comparable across courts.

A third contribution, by Palumbo et al. (2013) for the OECD, instead differs from the two just described, being more descriptive than analytical in nature, while still retaining significant informational value. This study places Spain at the OECD average in terms of overall length of civil proceedings (778 days compared to an average of 788), with times of 272 days at first instance, 189 days on appeal, and 316 days at the last instance, and a summary Doing Business measure of 515 days.

Overall, then, the existing literature on Spain focuses on specific contexts, without offering a comparative assessment of the efficiency of ordinary Spanish courts over a large sample and an extended time horizon.

3. Data and Methodology

3.1 Data

The empirical analysis is based on a panel dataset covering 80 *Juzgados de Instrucción* (Instruction Courts) distributed across the 17 autonomous communities of Spain, observed annually from 2016 to 2023. The courts analysed constitute the court of first instance of the Spanish criminal justice system, responsible for the investigation and preliminary instruction of criminal proceedings. The data were obtained from the official statistics published by the *Consejo General del Poder Judicial* (CGPJ), the governing body of the Spanish judiciary, which systematically collects operational and human resources data at the judicial district level.

Each district represents a decision-making unit (DMU) in the analysis. The sample comprises 640 court-year observations ($80 \text{ districts} \times 8 \text{ years}$), which generate seven consecutive period transitions for the calculation of the Malmquist Productivity Index. The chosen time span includes both the pre-pandemic years and the COVID-19 period (2019-2021), thus allowing us to capture the impact of the health emergency on judicial productivity.

The input and output variables are chosen in line with the literature discussed in Chapter 2. Four input variables were selected, all measured as counts in full-time equivalent units (FTE, *Full-Time Equivalent*): the number of judges (FTEJUDGES), the number of clerks (FTECLERKS), the number of case managers (FTEMAN) and the number of other support staff (FTEOTHER). These categories were chosen because they represent the main productive factors available to each court, namely the judicial staff, the administrative and legal support, the management staff and the auxiliary workers. Moreover, the use of FTE counts, rather than headcounts, ensures that part-time positions are appropriately weighted, providing a more accurate measure of the labour resources effectively employed. Indeed, a judge working at 50% does not contribute to the productive capacity of the court in the same way as a full-time colleague.

Similarly, the decision not to include quality variables, such as the appeal rate of judgments or the average time to disposition, reflects the limits of data availability at the district level. Indeed, in terms of variable selection for this type of study, as observed by several studies in the literature examined in Chapter 2, the quality of judicial decisions is difficult to capture through

aggregate quantitative indicators, to the point where its omission has become common practice in most studies in this field.

The output variable is instead represented by the number of resolved proceedings (RESOLVED), defined as the total number of criminal proceedings concluded in a given year. This is the most widely adopted output measure in the judicial efficiency literature, as it directly captures the primary function of the court, as discussed in Chapter 2. The dataset also contains information on pending proceedings (PENDING) and incoming cases (INCOMING), but these variables are not included as outputs. This is due to the fact that incoming cases reflect the exogenous volume of demand directed at each court, rather than its productive performance, while pending cases represent a stock variable whose annual variation may confuse the accumulation of backlogs with case complexity, complicating the interpretation of productivity changes. Resolved proceedings therefore remain the most internally consistent and comparable output measure across courts and years.

Table 3.1 reports the descriptive statistics for all variables, aggregated over the entire panel. Considerable heterogeneity can be noted among the various judicial districts. Indeed, we can observe a standard deviation of RESOLVED (27,614 proceedings) that almost equals its mean value (21,558), and a notable difference in cases resolved per year between the most active and the smallest court, which count respectively 234,000 and fewer than 5,000. A similar pattern emerges for the input variables, where the FTE counts for case managers and other support staff show coefficients of variation greater than one. In the face of such variability among courts, a flexible non-parametric method is more appropriate than approaches that assume a predefined production technology.

Table 3.1. Descriptive statistics (80 Instruction Courts, 2016-2023, N = 640)

Variable	Description	Mean	Std. Dev.	Min	Max
FTEJUDGES	Judges (FTE)	17.8	20.6	6.0	170.5
FTECLERKS	Clerks (FTE)	17.7	20.5	6.0	167.3

Variable	Description	Mean	Std. Dev.	Min	Max
FTEMAN	Case managers (FTE)	55.6	81.6	17.0	653.6
FTEOTHER	Other support staff (FTE)	104.2	126.9	29.3	1,033.9
RESOLVED	Resolved proceedings	21,558	27,614	4,028	234,424

Note: FTE = full-time equivalent. Statistics calculated on 640 court-year observations.

It can be noted that the heterogeneity is not distributed randomly across the territory. The courts of the large metropolitan areas, in particular those of the districts of Madrid and Barcelona, indeed tend to cluster in the upper tail of the distribution both for inputs and outputs, while the districts of the less populous autonomous communities are stably positioned in the lower tail. This geographical structure suggests that scale differences largely reflect the demographic density of the territory served, which represents an exogenous factor with respect to the organisational choices of individual courts. The regional analysis will be taken up again in Chapter 4, where it will be examined whether productivity gaps follow a systematic geographical pattern.

It is appropriate, however, to acknowledge some limitations of the dataset used. First, it should be noted that the data do not allow distinguishing the different complexity of proceedings; a simple theft case and an organised crime proceeding weigh the same in the output indicator, introducing a potential source of distortion in comparisons between courts with different workload compositions. Second, the CGPJ statistics are aggregated at the district level and do not allow observing the internal variability among individual judges operating in the same court. Finally, the observation period includes the COVID-19 pandemic, which introduced external shocks to judicial activity, such as hearing suspensions and registry closures, creating discontinuities that are not attributable to variations in organisational efficiency. This aspect will be explicitly analysed in Chapter 4, distinguishing the pre-COVID, COVID and post-COVID periods.

3.2 Methodology

The empirical analysis is based on the Malmquist Productivity Index (MPI), a non-parametric approach to measuring the variation in total factor productivity (TFP) over time. The MPI was originally formulated by Caves, Christensen and Diewert (1982) within the theory of index numbers. Färe, Grosskopf, Lindgren and Roos (1994) subsequently demonstrated that the index can be operationalised through distance functions estimated via *Data Envelopment Analysis* (DEA), making it applicable to contexts where output prices are absent or unreliable, a condition that characterises public entities such as courts.

As discussed in Chapter 2, DEA is a linear programming technique that constructs an empirical production frontier by enveloping the observed input-output combinations of all DMUs. When the analysis is output-oriented, the DEA distance function $D^o(x, y)$ for a given DMU with input vector x and output vector y measures the maximum proportional expansion of outputs that is achievable given the inputs and the prevailing technology. A value equal to one indicates that the DMU is on the production frontier, while values below one signal technical inefficiency. This type of formulation, that is output-oriented, is the most appropriate for the context here, as courts have, in the short run, a relatively fixed resource endowment, given that judges and staff cannot be quickly adjusted, so the real question is how many more proceedings could have been resolved with the same inputs.

Formally, consider a set of n DMUs, each of which produces s outputs using m inputs. For the reference DMU (DMU_0), the output-oriented efficiency score is obtained by solving the following linear programming problem:

$$\max \theta$$

subject to:

$$\sum_j \lambda_j x_{ij} \leq x_{i0}, \quad i = 1, \dots, m$$

$$\sum_j \lambda_j y_{rj} \geq \theta y_{r0}, \quad r = 1, \dots, s$$

$$\lambda_j \geq 0, \quad j = 1, \dots, n$$

where x_{ij} and y_{rj} are respectively the quantity of the i -th input and the r -th output of the j -th DMU, while λ_j are the weights indicating how much each reference DMU contributes to

defining the frontier point. The parameter $\theta \geq 1$ represents the proportional expansion factor of outputs: an optimal value $\theta^* = 1$ indicates that the DMU is on the production frontier and therefore technically efficient, while $\theta^* > 1$ indicates that the observed outputs could be proportionally increased by a factor θ^* while remaining within the production possibilities set. The output distance function is defined as $D^o(x, y) = 1/\theta^*$, so that a value equal to one corresponds to technical efficiency and a value below one signals inefficiency. Under the CRS assumption no constraint is added on the intensity of the weights; adding the constraint $\sum_j \lambda_j = 1$ yields instead the variable returns to scale (VRS) model of Banker, Charnes and Cooper (1984).

The calculation of the Malmquist index requires the estimation of four distance functions for each pair of consecutive periods $(t, t + 1)$. For each DMU it is necessary to solve four distinct linear programmes:

- (1) $D^{ot}(x^t, y^t)$: efficiency of the DMU in period t evaluated against the frontier of period t . The conventional DEA in the current period;
- (2) $D^{ot+1}(x^{t+1}, y^{t+1})$: efficiency in period $t + 1$ against the frontier of period $t + 1$. The conventional DEA in the following period;
- (3) $D^{ot}(x^{t+1}, y^{t+1})$: efficiency in period $t + 1$ against the frontier of period t . A DEA that compares future performance with past technology;
- (4) $D^{ot+1}(x^t, y^t)$: efficiency in period t against the frontier of period $t + 1$. A DEA that compares past performance with future technology.

Programmes (1) and (2) are conventional DEAs, whose results measure the efficiency of each DMU against the contemporary frontier. Programmes (3) and (4) are instead intertemporal, thus comparing each DMU with the frontier of a different period from the one in which it operates. Through these cross-comparisons it is possible to separate the efficiency change component (EC) from the technical change component (TC). Specifically, the EC component is determined entirely by programmes (1) and (2), while TC is calculated as the geometric mean of the ratios derived from (1), (2), (3) and (4). The corresponding values of MPI, EC and TC for each DMU and each period were calculated using R software, with which it is possible to solve the four linear programmes described above.

The output-oriented MPI between periods t and $t + 1$ for a given DMU is defined as the geometric mean of two period-specific ratios (Färe et al., 1994):

$$MPI = [M^t \times M^{t+1}]^{1/2}$$

where $M^t = D^{ot}(x^{t+1}, y^{t+1}) / D^{ot}(x^t, y^t)$ and $M^{t+1} = D^{ot+1}(x^{t+1}, y^{t+1}) / D^{ot+1}(x^t, y^t)$. Here $D^{os}(x, y)$ denotes the output distance function calculated with reference to the technology at period s . The geometric mean formulation avoids the arbitrariness of choosing as the sole reference either period t or $t + 1$, and the resulting index is independent of the choice of base period. An MPI value greater than one indicates a productivity improvement between the two periods; a value below one indicates a decline; a value equal to one denotes stagnation.

Following the decomposition proposed by Färe et al. (1994), the MPI can be factored into two economically interpretable components:

$$MPI = EC \times TC$$

The first component, EC (efficiency change, or *catch-up*), measures how much the DMU has moved, i.e. approached or moved away from, the contemporary production frontier between t and $t + 1$. Formally, $EC = D^{ot+1}(x^{t+1}, y^{t+1}) / D^{ot}(x^t, y^t)$. An EC value greater than one means that the DMU has reduced its distance from the frontier (catch-up); a value below one indicates regression. The second component, TC (technical change, or frontier shift), represents how much the best practice frontier has shifted between the two periods and is measured as the geometric mean of the ratios between the frontiers of the two periods, calculated using the input-output data of both periods. A TC value greater than one indicates that the frontier has expanded outward, meaning that the most efficient courts produce more output with the same inputs, while a value below one implies a contraction.

This decomposition is particularly informative in the judicial context. A productivity improvement driven mainly by a high EC would indeed suggest that the less efficient courts are registering improvements in terms of efficiency, catching up with the best performers and indicating margins for managerial or organisational improvement at the level of the individual court. On the contrary, an improvement driven by TC would signal more system-wide advances, such as, for example, the introduction of digital case management systems or changes to procedural rules, which raise the performance of all courts simultaneously. Distinguishing the

two components therefore informs judicial policies on where efficiency gains can be expected to originate.

Both the MPI and its components are calculated under the assumption of constant returns to scale (CRS). CRS is the standard assumption in the calculation of the Malmquist index and ensures that the resulting index satisfies the property of multiplicative consistency (Caves et al., 1982). Under the CRS assumption, the required distance functions are derived by solving the standard DEA linear programmes (Charnes, Cooper and Rhodes, 1978). Although in principle variable returns to scale (VRS) could also be used, the decomposition $MPI = EC \times TC$ is only valid under CRS; adopting VRS would introduce a third component and could make the index non-calculable for some observations (Färe et al., 1994). Moreover, the imposition of CRS is in line with the objective of measuring productivity change independently of scale differences between districts, given the strong size differences among courts documented in Table 3.1. The calculations are finally carried out in R using the *Benchmarking* package, which implements the *malmquist()* function for panel data (Bogetoft and Otto, 2011).

4. Results

In this chapter, the results of the Malmquist Productivity Index (MPI) calculation and its components, efficiency change (EC) and technical change (TC), are presented for the 80 *Juzgados de Instrucción* analysed over the period 2016-2023. The results are organised into five sections: the analysis by period, which examines the evolution of productivity across the seven consecutive annual transitions; the analysis by COVID phases, which compares the behaviour of the judicial system before, during and after the pandemic; the analysis by region, which identifies the geographical differences across the 17 autonomous communities; the analysis by district, which identifies the most and least productive courts and the diffusion of improvement; and finally the relationship between court size and productivity.

4.1 Analysis by period

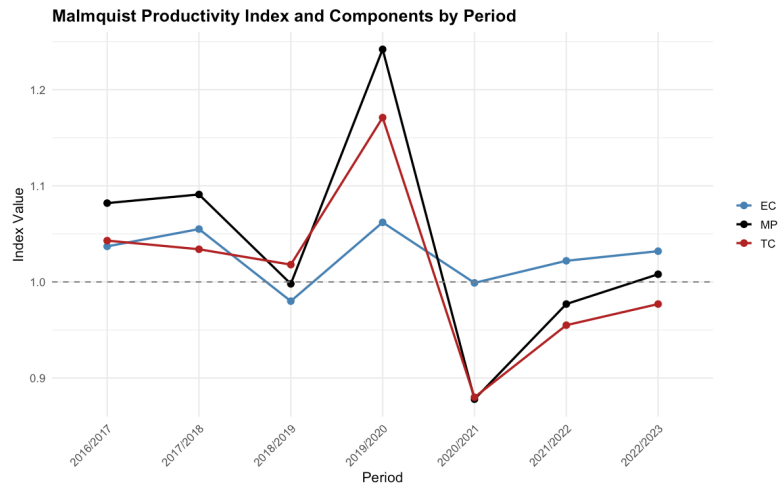
Table 4.1 reports the average values of MPI, EC and TC for each of the seven annual transitions considered. Figure 4.1 illustrates their evolution over time, highlighting the demarcation line between productivity growth and decline (value 1).

Table 4.1. *Malmquist Index and components by period (averages across 80 courts)*

Period	MPI	EC	TC
2016/2017	1.082	1.037	1.043
2017/2018	1.091	1.055	1.034
2018/2019	0.998	0.980	1.018
2019/2020	1.242	1.062	1.171
2020/2021	0.878	0.999	0.880
2021/2022	0.977	1.022	0.955
2022/2023	1.008	1.032	0.977

Note: Values above 1 indicate improvement; values below 1 indicate deterioration.

Figure 4.1. Evolution of MPI, EC and TC by period (2016-2023)



Overall, the Spanish judicial system recorded productivity improvements in the first two periods considered: MPI = 1.082 in 2016/2017 and MPI = 1.091 in 2017/2018. In both cases, both EC and TC contribute positively, showing how the courts not only recovered efficiency relative to the frontier, but that the frontier itself shifted outward.

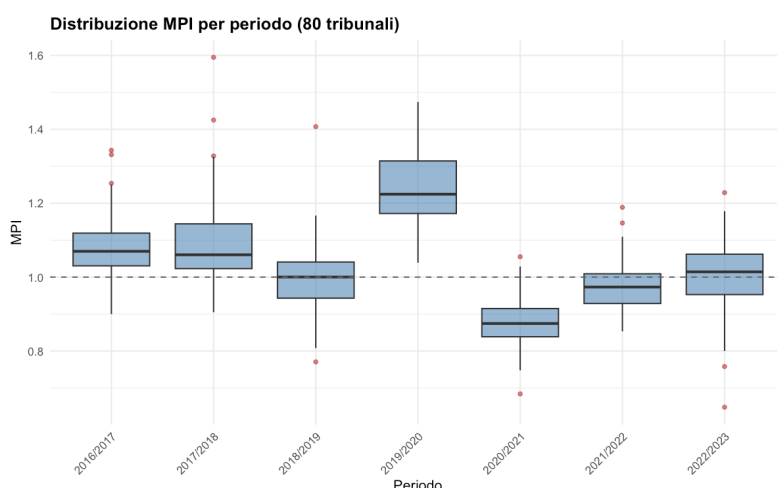
The period 2018/2019 marks a first interruption: MPI falls to 0.998, broadly in line with stagnation. In this case, the EC component falls below one (0.980), indicating a slight distancing from the frontier for the average court, while TC remains slightly positive (1.018). This signals that the frontier continued to advance, but most courts were unable to keep pace.

The most evident peak is recorded in the period 2019/2020, with MPI = 1.242. This result, which may appear positive, should be interpreted with caution, as it could mainly reflect the impact of the COVID-19 pandemic, which drastically reduced the number of hearings and incoming proceedings in the second half of 2020, while resolved proceedings, calculated on a calendar year basis, experienced a less immediate contraction. The strong contribution of TC (1.171) suggests a shift in the frontier linked not to an organisational improvement, but to the pandemic shock on the workload.

The subsequent transition, 2020/2021, shows the sharpest decline of the entire period: $MPI = 0.878$. In this case, the EC component is essentially stationary (0.999), while the collapse is entirely attributable to $TC = 0.880$. This indicates a contraction of the production frontier, in line with a period in which courts were trying to resume functioning despite the difficult conditions.

In the post-pandemic periods (2021/2022 and 2022/2023), productivity remains below or just above the unit (0.977 and 1.008 respectively), with TC struggling to recover to pre-COVID levels. EC, on the other hand, remains positive in both periods, suggesting that courts are gradually recovering individual efficiency, but that the systemic frontier has not yet returned to previous levels.

Figure 4.2. Distribution of MPI by period (80 courts)



The boxplot in Figure 4.2 offers a complementary view, showing the distribution of individual MPI values for each period beyond the averages alone. In the first two periods, the distributions are concentrated above the unit, with medians around 1.08 and some upward outliers. The period 2018/2019 shows a balanced distribution around the parity threshold, with the median falling to almost exactly 1.

The period 2019/2020 presents the highest and most dispersed distribution of the entire sample: the median exceeds 1.20 and outliers up to 1.6 are recorded. As already discussed, this reflects the distortion induced by the pandemic. The sharp decline is observed in 2020/2021, with the distribution compact and clearly below the unit (median ≈ 0.87). In the last two periods, the

distributions gradually rise towards the unit threshold, with greater dispersion in 2022/2023, indicating a recovery that is still heterogeneous across courts.

4.2 Analysis by COVID phases

In order to isolate the effect of the pandemic on judicial activity, the seven periods were grouped into three phases: pre-COVID (2016/2017, 2017/2018 and 2018/2019), COVID (2019/2020) and post-COVID (2020/2021, 2021/2022 and 2022/2023). Table 4.2 reports the average values of MPI, EC and TC for each phase.

Table 4.2. *Malmquist Index and components by COVID phase (averages across 80 courts)*

Phase	MPI	EC	TC
Pre-COVID	1.057	1.024	1.032
COVID	1.242	1.062	1.171
Post-COVID	0.954	1.018	0.937

Note: Average values by phase. Pre-COVID: periods 2016/17-2018/19; COVID: 2019/20; Post-COVID: 2020/21-2022/23.

In the pre-COVID period, average annual productivity was 1.057, with both components positive: EC = 1.024 and TC = 1.032. This indicates that, before the pandemic, the Spanish judicial system was improving both in terms of individual court efficiency and in terms of overall technological and organisational advancement.

The COVID period records the highest MPI value (1.242), driven mainly by TC = 1.171. As already discussed in the analysis by period, this result should be interpreted with caution: the strong shift in the frontier largely reflects the distortion introduced by the pandemic in the relationship between inputs and outputs, rather than a real advancement in system efficiency.

The post-COVID period is the most concerning. MPI = 0.954 indicates an average decline in productivity. The TC component (0.937) is responsible for the decline, while EC remains positive (1.018), confirming that courts are individually recovering relative to the frontier, but

that the frontier itself has contracted. This picture suggests that the impact of the pandemic has left a lasting mark on the overall productive capacity of the system, at least within the time horizon analysed.

Figure 4.3. EC and TC components by COVID phase

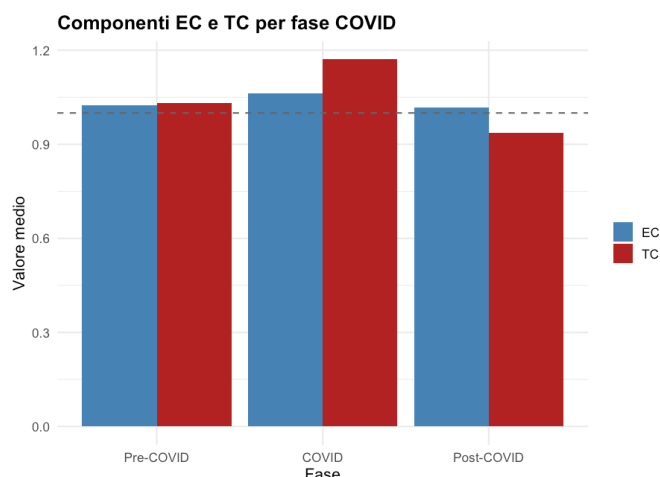


Figure 4.3 breaks down the EC and TC components by phase, so as to observe the source of the productivity change. In the pre-COVID period, EC and TC are almost equivalent (1.024 and 1.032 respectively), indicating a balanced improvement between individual efficiency and frontier advancement. During the COVID phase, TC reaches its highest value (1.171), while EC, though positive, remains well below it. In the post-COVID period the situation reverses. Indeed, EC still exceeds the unit (1.018) while TC falls to 0.937, confirming that courts recover individual efficiency in a context of contracting frontier.

4.3 Analysis by region

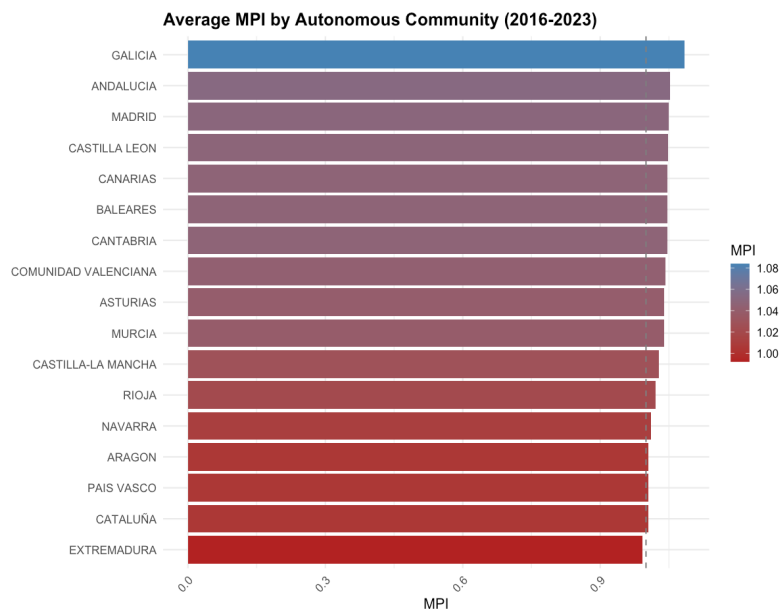
Table 4.3 and Figure 4.4 present the average values of MPI, EC and TC for each of the 17 autonomous communities, calculated over the entire period 2016-2023. The regional analysis allows us to verify whether the productivity gaps identified at the aggregate level follow a systematic geographical pattern.

Table 4.3. *Malmquist Index and components by autonomous community (averages 2016-2023)*

Autonomous community	MPI	EC	TC
Galicia	1.084	1.063	1.016
Andalucía	1.053	1.042	1.009
Madrid	1.050	1.042	1.006
Castilla y León	1.048	1.041	1.008
Cantabria	1.047	1.037	1.009
Baleares	1.047	1.028	1.016
Canarias	1.047	1.029	1.019
Comunidad Valenciana	1.043	1.031	1.011
Asturias	1.040	1.027	1.011
Murcia	1.039	1.027	1.012
Castilla-La Mancha	1.028	1.023	1.005
Rioja	1.021	0.990	1.031
Navarra	1.011	1.009	1.006
Cataluña	1.005	0.993	1.011
País Vasco	1.005	0.999	1.005
Aragón	1.005	0.993	1.007
Extremadura	0.992	0.978	1.009

Note: Autonomous communities ordered by decreasing MPI. Average values calculated across the seven annual transitions.

Figure 4.4. Average MPI by autonomous community (2016-2023)



The results highlight significant geographical variability. Galicia records the highest MPI value (1.084), followed by Andalucía (1.053) and Madrid (1.050). In all three communities, improvement is driven mainly by EC, indicating that courts in these regions have on average reduced their distance from the production frontier over the period analysed.

At the opposite extreme, Extremadura is the only autonomous community with MPI below one (0.992), indicating an average decline in productivity over the time period considered. In this case too, the EC component is negative (0.978), confirming a distancing from the frontier for the courts of this region. Aragón, Cataluña and País Vasco are positioned just above the unit threshold, with almost negligible productivity changes.

An interesting observation concerns the Rioja, which despite having an EC below one (0.990), records the highest TC among all autonomous communities (1.031). This indicates that the courts of Rioja benefited from advances in the production frontier, that is, an improvement of the overall system, but without managing to keep pace individually, resulting relatively less efficient compared to the others.

Overall, no clear north-south or centre-periphery pattern emerges. The communities with the best results include both large and populous regions such as Andalucía and Madrid, and smaller regions such as Galicia and Cantabria. This suggests that productivity differences are not simply

a function of the size of the regional judicial system, but reflect organisational and structural factors specific to each community.

4.4 The most and least productive courts

Table 4.4 reports the five courts with the highest average MPI value and the five with the lowest, calculated over the entire period 2016-2023.

Table 4.4. Top 5 and Bottom 5 courts by average MPI (2016-2023)

District	MPI	EC	TC
Top 5			
Pontevedra	1.122	1.095	1.016
Lugo	1.112	1.088	1.020
Jaén	1.107	1.098	1.008
Cádiz	1.100	1.075	1.018
Granada	1.098	1.085	1.008
Bottom 5			
Badalona	0.990	0.974	1.018
Lleida	0.988	0.982	1.005
Tarragona	0.986	0.977	1.010
Badajoz	0.954	0.947	1.007
Donostia-San Sebastián	0.952	0.946	1.005

Note: Average values calculated across the seven annual transitions (2016-2023).

The courts with the greatest productivity growth are Pontevedra (MPI = 1.122) and Lugo (1.112), both belonging to Galicia, the autonomous community with the best average result at

the regional level. Three Andalusian courts follow, namely Jaén (1.107), Cádiz (1.100) and Granada (1.098), confirming the good positioning of that region in the aggregate ranking. In all five cases, improvement is mainly due to EC, with values ranging between 1.075 and 1.098.

At the opposite extreme, the least productive courts are Donostia-San Sebastián (MPI = 0.952) and Badajoz (0.954), followed by Tarragona (0.986), Lleida (0.988) and Badalona (0.990). In the first two cases, EC is significantly below the unit. That's indicate a distancing from the production frontier over the period analysed. It is worth noting that three of the five least productive courts belong to Cataluña (Tarragona, Lleida and Badalona), which helps explain the poor positioning of that community in the regional ranking.

Table 4.5. *Share of courts with MPI > 1 by period*

Period	% of courts with MPI > 1
2016/2017	90.0%
2017/2018	82.5%
2018/2019	51.2%
2019/2020	100.0%
2020/2021	3.8%
2021/2022	28.7%
2022/2023	56.2%

Note: The share indicates the percentage of courts (out of 80) with average MPI above one in the period considered.

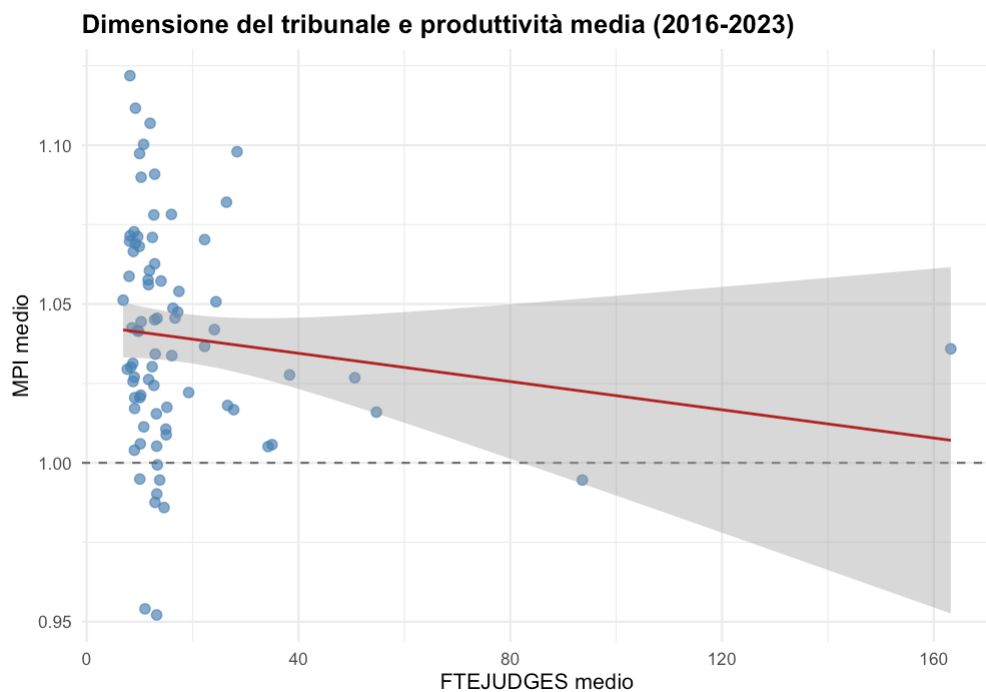
The data in Table 4.5 clearly confirm the impact of the pandemic. In the period 2019/2020 all 80 courts recorded MPI > 1 (100%), a result that, as already stated, reflects the distortion induced by the reduction in incoming proceedings rather than a real systemic improvement. Conversely, in the period 2020/2021 only 3.8% of courts recorded productivity growth: the lowest figure in the entire sample.

In the two pre-COVID periods with positive average MPI (2016/2017 and 2017/2018), the share of courts with $MPI > 1$ was 90.0% and 82.5% respectively, confirming that improvement was widely diffused and not limited to a few units. In the most recent period (2022/2023), the share rises to 56.2%, a sign of a recovery that is still incomplete but progressing relative to the minimum of 2020/2021.

4.5 Court size and productivity

Figure 4.5 relates the average size of each court, measured by the average number of full-time equivalent judges (FTEJUDGES) over the period 2016-2023, to its average MPI over the same time horizon.

Figure 4.5. Court size and average productivity (2016-2023)



From the figure it can be seen that the regression shows a negative slope, highlighting how courts of larger size tend to have slightly lower productivity levels in the period considered. However, the confidence interval is very wide, especially for the largest courts, where the data are limited, and consequently the relationship is weaker.

Finally, most courts are concentrated in the range with fewer than 20 FTEJUDGES, showing considerable MPI variability for a given size. The largest courts ($FTEJUDGES > 80$) are few and tend to be positioned close to the unit threshold, without showing exceptional growth or

decline in productivity. Overall, the results suggest that court size alone is not a significant determinant of productivity change in the period analysed, and that other factors, such as organisational, territorial, or those related to the composition of the workload, play a more relevant role.

5. Conclusions

The aim of this work was to analyse the productivity change of the 80 *Juzgados de Instrucción* in Spain over the period 2016-2023 through the Malmquist Productivity Index (MPI). The analysis was carried out across 17 autonomous communities for seven consecutive annual transitions, and the results were examined along five dimensions: the evolution over time, the impact of the COVID-19 pandemic, the geographical differences between autonomous communities, the differences between individual districts, and the relationship between court size and productivity.

Starting from the evolution over time, the results obtained are rather heterogeneous. In the first two periods analysed (2016/2017 and 2017/2018), the system recorded productivity growth, with MPI equal to 1.082 and 1.091 respectively. In both cases, the improvement was driven by both EC and TC, indicating a balanced advancement between individual efficiency recovery and a shift of the production frontier. The first setback, however, comes in the period 2018/2019, with MPI essentially equal to one (0.998) and a slight contraction in EC, indicating that most courts were unable to keep pace with a frontier that nonetheless continued to advance.

The period 2019/2020 recorded the highest MPI value in the entire sample (1.242), driven mainly by TC = 1.171. As discussed in Chapter 4, this result should be interpreted carefully, as the distortion created by the pandemic plays a key role in the frontier shift: incoming proceedings experienced a drastic reduction while resolved proceedings remained relatively stable in the short term. The improvement is therefore partly artificial, linked to the disruption of the relationship between inputs and outputs rather than to a real organisational advancement of the system.

The transition 2020/2021 showed the sharpest decline, with MPI = 0.878, attributable almost entirely to the contraction of TC (0.880). The EC component remained essentially stationary (0.999), which may be more related to a systemic contraction of the frontier than to a decline in the efficiency of individual courts. In the subsequent periods (2021/2022 and 2022/2023), productivity showed signs of gradual recovery. Specifically, EC turned positive in both periods, while TC continued to struggle to recover to pre-COVID levels. The impact of the pandemic therefore appears still visible within the time horizon analysed, particularly in the technological component.

The analysis by COVID phases effectively summarised the picture just described. In the pre-COVID phase (2016/2017-2018/2019), the average MPI was 1.057, with both components positive. In the COVID period (2019/2020), MPI reached 1.242, with an exceptionally high TC. In the post-COVID phase (2020/2021-2022/2023), MPI fell to 0.954, with TC = 0.937 responsible for the overall decline. The share of courts with MPI > 1 dropped from 100% in 2019/2020 to 3.8% in 2020/2021, confirming the near-universal nature of both the artificial peak and the subsequent collapse.

The results obtained fit into the broader framework of the literature on the efficiency of Spanish courts. Pedraja-Chaparro and Salinas-Jiménez (1996) had found an average efficiency of 77.38% for the High Courts of Justice in 1991, with only five out of twenty-one courts classified as technically efficient. Espasa and Esteller-Moré (2015), analysing Catalan civil courts over the period 2005-2013, found instead an average efficiency of 97.82%, increasing over time, suggesting a significant improvement in the long run. Although direct comparison with these studies is limited by differences in the type of courts analysed and the method employed, the overall picture emerging from the literature is that of a Spanish judicial system that has shown a tendency towards improvement over the long term, but with persistent differences across geographical areas and jurisdictions. The results of this work fit into this trajectory, confirming a positive dynamic in the pre-COVID period and identifying the pandemic as a breaking point that interrupted, at least temporarily, the path of improvement.

On the geographical level, the results highlight significant variability across the 17 autonomous communities. The highest average MPI was recorded by Galicia, with an MPI of 1.084, followed by Andalucía with 1.053 and Madrid with 1.050. Extremadura is the only community with an average MPI below one, at 0.992. Improvement across almost all regions was driven mainly by EC, with TC contributing to a lesser extent. An anomalous case is that of Rioja, which records the highest TC among all autonomous communities (1.031) despite an EC below the unit, which shows that the frontier advanced but the courts of that region were unable to keep pace individually. No clear north-south or centre-periphery geographical pattern emerged, suggesting that productivity differences reflect factors specific to each community rather than a purely territorial logic.

At the level of individual districts, the most productive courts belong mainly to Galicia (Pontevedra, MPI = 1.122; Lugo, MPI = 1.112) and Andalucía (Jaén, Cádiz, Granada), in line

with the good positioning of these regions in the aggregate ranking. The least productive include three courts in Cataluña (Badalona, Lleida, Tarragona) and Donostia-San Sebastián, the district with the lowest value (0.952). A significant finding concerns the level of EC, which in almost all extreme cases proves to be the most discriminating component: the most productive courts show high EC values, while the least productive present EC below the unit. The relationship between court size and average MPI is instead weakly negative but statistically not very robust, suggesting that economies or diseconomies of scale are not the main determining factor of productivity change in the period analysed.

In terms of policy implications, the results offer several indications. The fact that EC was the main component of improvement in the pre-COVID period suggests that there is room to recover individual efficiency within the system, without necessarily intervening on the structural and organisational factors that determine the position of the frontier. At the same time, the contraction of TC in the post-COVID period signals that systemic improvement is difficult to maintain in the presence of relevant exogenous shocks. Consequently, policies aimed at improving court efficiency should act on both dimensions: on one hand, supporting efficiency recovery in the most lagging courts through targeted interventions at the level of the individual district; on the other, investing in organisational and technological reforms capable of shifting the production frontier in a stable way over time, thus acting on the system as a whole.

A second relevant aspect for policy concerns the geographical heterogeneity identified. The fact that Galicia and Andalucía are consistently among the most productive regions, while Extremadura and some Catalan courts record below-average performance, suggests that efficiency policies should not be applied uniformly across the national territory. Targeted interventions at the regional or district level, taking into account local specificities in terms of workload, available resources and internal organisation, seem more appropriate than generalised reforms. In this sense, a comparative analysis between districts such as the one proposed in this work can serve as a useful tool for institutions in identifying the most critical contexts and allocating available resources.

Finally, the sharp contraction in productivity in the period 2020/2021, with only 3.8% of courts able to record a net improvement, highlights the vulnerability of the judicial system to large exogenous shocks. This suggests that, in addition to ordinary efficiency policies, it would be

advisable to put in place crisis management mechanisms within the judicial organisation, such as business continuity plans, procedures for holding hearings remotely and systems for redistributing workload between districts in the event of an emergency. Such tools would not eliminate the effects of a shock like the pandemic, but could reduce its impact on productivity and accelerate recovery in the subsequent phase.

There are, however, some limitations in this study that should be made explicit. First, the output variable used, the number of resolved proceedings, is an aggregate measure that does not distinguish between different types of proceedings varying in complexity and duration. As shown by Ippoliti and Tria (2020) in the Italian context, disaggregating the output can significantly change efficiency estimates and reduce the apparent gap between different geographical areas. Second, no quality variable for decisions is included in the model, as such measures are not available at the district level. The variable most commonly used for this purpose in the literature is the appeal rate of rulings, which however was not available for the sample analysed. Finally, the model does not account for external contextual variables, such as the crime rate, the population of the district or the composition of the workload by type of offence, which could influence measured productivity independently of court efficiency.

Possible future developments include the introduction of a second-stage analysis, in which the estimated MPI, EC and TC values are regressed on contextual variables in order to identify their determinants, following the two-stage approach common in the DEA literature. It would also be interesting to extend the analysis to other types of Spanish courts, such as the *Juzgados de lo Social* or the *Juzgados de Primera Instancia*, to verify whether the productivity patterns found for the *Juzgados de Instrucción* are specific to this type or reflect more general dynamics of the judicial system. A temporal extension including more recent data would also allow a more complete assessment of the system's ability to recover pre-pandemic productivity levels.

Overall, this work shows that the system of Spanish *Juzgados de Instrucción* went through a period of significant productivity variability between 2016 and 2023, with a clear breaking point linked to the COVID-19 pandemic. The MPI-based approach made it possible to distinguish between different components of improvement and to identify geographical patterns that would not have emerged from a purely aggregate or static efficiency analysis. It is hoped that this type of analysis can contribute, alongside other complementary approaches, to a more systematic assessment of judicial efficiency in Spain and to informing more evidence-based policy choices.

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